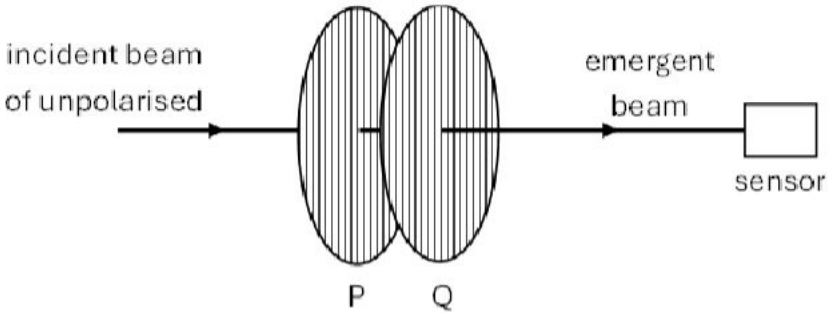
12 Two polarising filters, P and Q, are placed in front of a light sensor. Their transmission axes are initially parallel. Unpolarised light passes through them and the sensor records an intensity I.



Filter Q is then rotated while filter P remains fixed.
Which angle through which Q may be rotated would reduce the detected intensity by 36%?

A 23° **B** 37° **C** 50° **D** 53°

Ans: B

$I = I_0 \cos^2 \theta$

Since the detected intensity is reduced by 36%, 64% of the intensity is transmitted.
$I = 0.64 I_0$

$\cos^2 \theta = 0.64$

$\theta = 36.9^\circ$